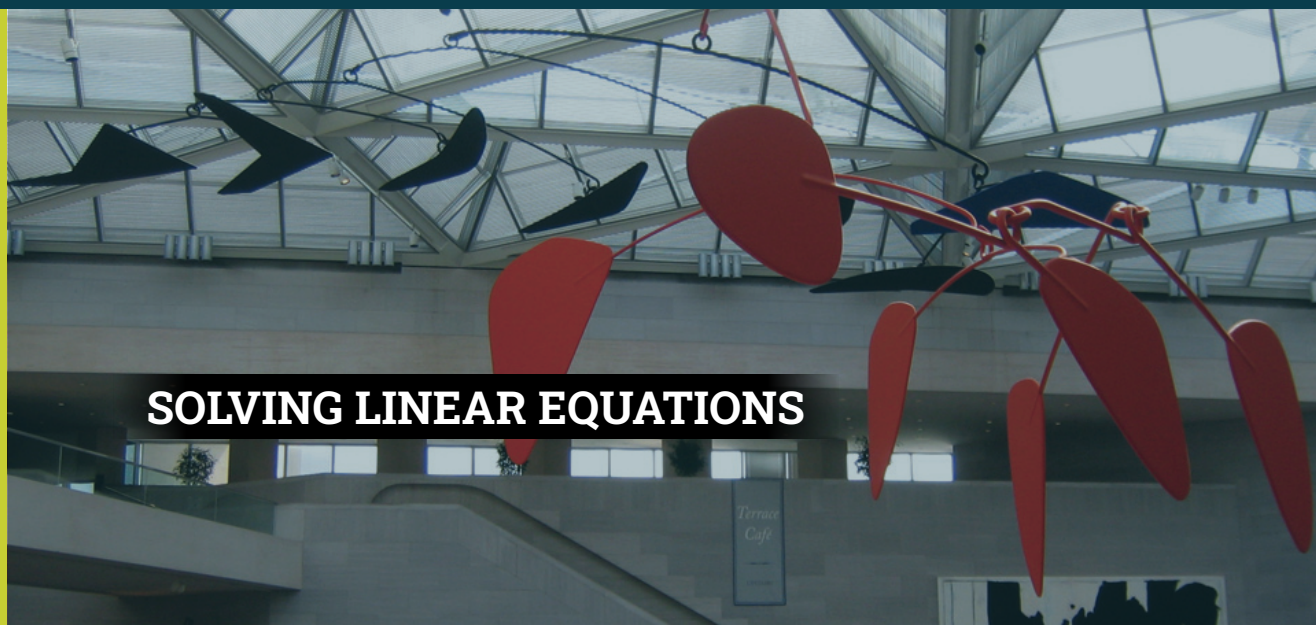


8 SOLVING LINEAR EQUATIONS

Figure 8.1 A Calder mobile is balanced and has several elements on each side. (credit: paurian, Flickr)

Chapter Outline

8.1 Solve Equations Using the Subtraction and Addition Properties of Equality
8.2 Solve Equations Using the Division and Multiplication Properties of Equality
8.3 Solve Equations with Variables and Constants on Both Sides
8.4 Solve Equations with Fraction or Decimal Coefficients

Introduction to Solving Linear Equations

Teetering high above the floor, this amazing mobile remains aloft thanks to its carefully balanced mass. Any shift in either direction could cause the mobile to become lopsided, or even crash downward. In this chapter, we will solve equations by keeping quantities on both sides of an equal sign in perfect balance.

8.1 Solve Equations Using the Subtraction and Addition Properties of Equality

Learning Objectives

By the end of this section, you will be able to:

- Solve equations using the Subtraction and Addition Properties of Equality
- Solve equations that need to be simplified
- Translate an equation and solve
- Translate and solve applications

BE PREPARED 8.1 Before you get started, take this readiness quiz.

Solve: $n - 12 = 16$.
If you missed this problem, review Example 2.33.

BE PREPARED 8.2 Translate into algebra 'five less than x.'
If you missed this problem, review Example 2.24.

BE PREPARED 8.3 Is $x = 2$ a solution to $5x - 3 = 7$?
If you missed this problem, review Example 2.28.

We are now ready to "get to the good stuff." You have the basics down and are ready to begin one of the most important

topics in algebra: solving equations. The applications are limitless and extend to all careers and fields. Also, the skills and techniques you learn here will help improve your critical thinking and problem-solving skills. This is a great benefit of studying mathematics and will be useful in your life in ways you may not see right now.

Solve Equations Using the Subtraction and Addition Properties of Equality

We began our work solving equations in previous chapters. It has been a while since we have seen an equation, so we will review some of the key concepts before we go any further.

We said that solving an equation is like discovering the answer to a puzzle. The purpose in solving an equation is to find the value or values of the variable that make each side of the equation the same. Any value of the variable that makes the equation true is called a solution to the equation. It is the answer to the puzzle.

Solution of an Equation

A **solution of an equation** is a value of a variable that makes a true statement when substituted into the equation.

In the earlier sections, we listed the steps to determine if a value is a solution. We restate them here.



HOW TO

Determine whether a number is a solution to an equation.

Step 1. Substitute the number for the variable in the equation.
Step 2. Simplify the expressions on both sides of the equation.
Step 3. Determine whether the resulting equation is true.
- If it is true, the number is a solution.
- If it is not true, the number is not a solution.

EXAMPLE 8.1

Determine whether $y = \frac{3}{4}$ is a solution for $4y + 3 = 8y$.

Solution

	$4y + 3 = 8y$
Substitute $\frac{3}{4}$ for y.	$4\left(\frac{3}{4}\right) + 3 \stackrel{?}{=} 8\left(\frac{3}{4}\right)$
Multiply.	$3 + 3 \stackrel{?}{=} 6$
Add.	$6 = 6$ ✓

Since $y = \frac{3}{4}$ results in a true equation, $\frac{3}{4}$ is a solution to the equation $4y + 3 = 8y$.

TRY IT 8.1 Is $y = \frac{2}{3}$ a solution for $9y + 2 = 6y$?

TRY IT 8.2 Is $y = \frac{2}{5}$ a solution for $5y - 3 = 10y$?

We introduced the Subtraction and Addition Properties of Equality in Solving Equations Using the Subtraction and Addition Properties of Equality. In that section, we modeled how these properties work and then applied them to solving

equations with whole numbers. We used these properties again each time we introduced a new system of numbers. Let's review those properties here.

Subtraction and Addition Properties of Equality

Subtraction Property of Equality

For all real numbers a, b, and c, if $a = b$, then $a - c = b - c$.

Addition Property of Equality

For all real numbers a, b, and c, if $a = b$, then $a + c = b + c$.

When you add or subtract the same quantity from both sides of an equation, you still have equality.

We introduced the Subtraction Property of Equality earlier by modeling equations with envelopes and counters. Figure 8.2 models the equation $x + 3 = 8$.

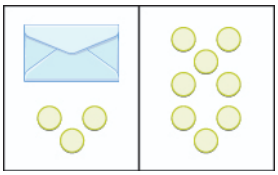


Figure 8.2

The goal is to isolate the variable on one side of the equation. So we 'took away' 3 from both sides of the equation and found the solution $x = 5$.

Some people picture a balance scale, as in Figure 8.3, when they solve equations.

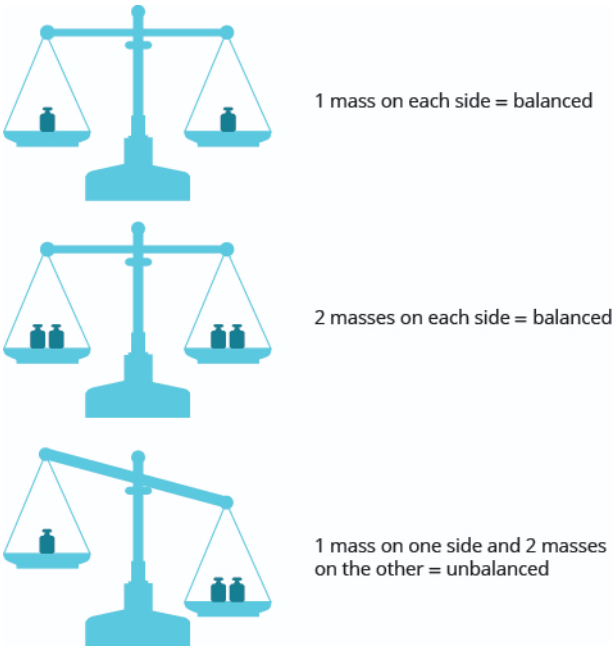


Figure 8.3

The quantities on both sides of the equal sign in an equation are equal, or balanced. Just as with the balance scale, whatever you do to one side of the equation you must also do to the other to keep it balanced.

Let's review how to use Subtraction and Addition Properties of Equality to solve equations. We need to isolate the variable on one side of the equation. And we check our solutions by substituting the value into the equation to make sure we have a true statement.

EXAMPLE 8.2

Solve: $x - 11 = -3$.

Solution

To isolate x, we undo the addition of 11 by using the Subtraction Property of Equality.

	$x - 11 = -3$
We "undo" the subtraction of 11 by adding 11 to each side.	$x - 11 + 11 = -3 + 11$
Simplify.	$x = 8$
Check:	$x - 11 = -3$
Substitute $x = 8$.	$8 - 11 \stackrel{?}{=} -3$
	$-3 = -3$ ✓

Since $x = 8$ makes $x - 11 = -3$ a true statement, we know that it is a solution to the equation.

TRY IT 8.3 Solve: $x + 9 = -7$.

TRY IT 8.4 Solve: $x + 16 = -4$.

In the original equation in the previous example, 11 was added to the x, so we subtracted 11 to 'undo' the addition. In the next example, we will need to 'undo' subtraction by using the Addition Property of Equality.

EXAMPLE 8.3

Solve: $m + 4 = -5$.

Solution

		$m + 4 = -5$
Subtract 4 from each side to "undo" the addition.		$m + 4 - 4 = -5 - 4$
Simplify.		$m = -9$
Check:	$m + 4 = -5$	
Substitute $m = -9$.	$-9 + 4 \stackrel{?}{=} -5$	
	$-5 = -5$ ✓	
		The solution to $m - 4 = -5$ is $m = -1$.

TRY IT 8.5 Solve: $n - 6 = -7$.

TRY IT 8.6 Solve: $x - 5 = -9$.

Now let's review solving equations with fractions.

EXAMPLE 8.4

Solve: $n - \frac{3}{8} = \frac{1}{2}$.

Solution

	$n - \frac{3}{8} = \frac{1}{2}$
Use the Addition Property of Equality.	$n - \frac{3}{8} + \frac{3}{8} = \frac{1}{2} + \frac{3}{8}$
Find the LCD to add the fractions on the right.	$n - \frac{3}{8} + \frac{3}{8} = \frac{4}{8} + \frac{3}{8}$
Simplify	$n = \frac{7}{8}$
Check:	$n - \frac{3}{8} = \frac{1}{2}$
Substitute $n = \frac{7}{8}$.	$\frac{7}{8} - \frac{3}{8} \stackrel{?}{=} \frac{1}{2}$
Subtract.	$\frac{4}{8} \stackrel{?}{=} \frac{1}{2}$
Simplify.	$\frac{1}{2} = \frac{1}{2}$ ✓
The solution checks.	

TRY IT 8.7 Solve: $p - \frac{1}{3} = \frac{5}{6}$.

TRY IT 8.8 Solve: $q - \frac{1}{2} = \frac{1}{6}$.

In Solve Equations with Decimals, we solved equations that contained decimals. We'll review this next.

EXAMPLE 8.5

Solve $a - 3.7 = 4.3$.

Solution

	$a - 3.7 = 4.3$
Use the Addition Property of Equality.	$a - 3.7 + 3.7 = 4.3 + 3.7$

Add.		$a = 8$
Check:	$a - 3.7 = 4.3$	
Substitute $a = 8$.	$8 - 3.7 \stackrel{?}{=} 4.3$	
Simplify.	$4.3 = 4.3$ ✓	
The solution checks.		

TRY IT 8.9 Solve: $b - 2.8 = 3.6$.

TRY IT 8.10 Solve: $c - 6.9 = 7.1$.

Solve Equations That Need to Be Simplified

In the examples up to this point, we have been able to isolate the variable with just one operation. Many of the equations we encounter in algebra will take more steps to solve. Usually, we will need to simplify one or both sides of an equation before using the Subtraction or Addition Properties of Equality. You should always simplify as much as possible before trying to isolate the variable.

EXAMPLE 8.6

Solve: $3x - 7 - 2x - 4 = 1$.

Solution

The left side of the equation has an expression that we should simplify before trying to isolate the variable.

	$3x - 7 - 2x - 4 = 1$
Rearrange the terms, using the Commutative Property of Addition.	$3x - 2x - 7 - 4 = 1$
Combine like terms.	$x - 11 = 1$
Add 11 to both sides to isolate x.	$x - 11 + 11 = 1 + 11$
Simplify.	$x = 12$

Check.
Substitute $x = 12$ into the original equation.

$$\begin{aligned} 3x - 7 - 2x - 4 &= 1 \\ 3(12) - 7 - 2(12) - 4 &= 1 \\ 36 - 7 - 24 - 4 &= 1 \\ 29 - 24 - 4 &= 1 \\ 5 - 4 &= 1 \\ 1 &= 1 \checkmark \end{aligned}$$

The solution checks.

TRY IT 8.11 Solve: $8y - 4 - 7y - 7 = 4$.

TRY IT 8.12 Solve: $6z + 5 - 5z - 4 = 3$.

EXAMPLE 8.7

Solve: $3(n - 4) - 2n = -3$.

Solution

The left side of the equation has an expression that we should simplify.

	$3(n-4)-2n=-3$
Distribute on the left.	$3n-12-2n=-3$
Use the Commutative Property to rearrange terms.	$3n-2n-12=-3$
Combine like terms.	$n-12=-3$
Isolate n using the Addition Property of Equality.	$n-12+12=-3+12$
Simplify.	$n=9$

Check.

Substitute $n = 9$ into the original equation.

$$\begin{aligned} 3(n-4)-2n &= -3 \\ 3(9-4)-2\cdot 9 &= -3 \\ 3(5)-18 &= -3 \\ 15-18 &= -3 \\ -3 &= -3 \checkmark \end{aligned}$$

The solution checks.

TRY IT 8.13 Solve: $5(p - 3) - 4p = -10$.

TRY IT 8.14 Solve: $4(q + 2) - 3q = -8$.

EXAMPLE 8.8

Solve: $2(3k - 1) - 5k = -2 - 7$.

Solution

Both sides of the equation have expressions that we should simplify before we isolate the variable.

	$2(3k-1)-5k=-2-7$
Distribute on the left, subtract on the right.	$6k-2-5k=-9$
Use the Commutative Property of Addition.	$6k-5k-2=-9$
Combine like terms.	$k-2=-9$

Undo subtraction by using the Addition Property of Equality.	$k-2+2=-9+2$
Simplify.	$k=-7$
Check. Let $k=-7$. $2(3k-1)-5k=-2-7$ $2(3(-7)-1)-5(-7)=-2-7$ $2(-21-1)-5(-7)=-9$ $2(-22)+35=-9$ $-44+35=-9$ $-9=-9$ ✓ The solution checks.	

TRY IT 8.15 Solve: $4(2h-3)-7h=-6-7$.

TRY IT 8.16 Solve: $2(5x+2)-9x=-2+7$.

Translate an Equation and Solve

In previous chapters, we translated word sentences into equations. The first step is to look for the word (or words) that translate(s) to the equal sign. Table 8.1 reminds us of some of the words that translate to the equal sign.

Equals (=)						
is	is equal to	is the same as	the result is	gives	was	will be

Table 8.1

Let's review the steps we used to translate a sentence into an equation.

HOW TO

Translate a word sentence to an algebraic equation.

Step 1. Locate the "equals" word(s). Translate to an equal sign.
Step 2. Translate the words to the left of the "equals" word(s) into an algebraic expression.
Step 3. Translate the words to the right of the "equals" word(s) into an algebraic expression.

Now we are ready to try an example.

EXAMPLE 8.9

Translate and solve: five more than x is equal to 26.

Solution

Translate.	$\underbrace{x+5}_{\text{Five more than } x} \quad \underbrace{=}_{\text{is equal to}} \quad \underbrace{26}_{26}$
Subtract 5 from both sides.	$x+5-5=26-5$
Simplify.	$x=21$
Check: Is 26 five more than 21? $21+5 \stackrel{?}{=} 26$ $26=26$ ✓ The solution checks.	

TRY IT 8.17 Translate and solve: Eleven more than x is equal to 41.

TRY IT 8.18 Translate and solve: Twelve less than y is equal to 51.

EXAMPLE 8.10

Translate and solve: The difference of $5p$ and $4p$ is 23.

Solution

Translate.	$\underbrace{5p-4p}_{\text{The difference of } 5p \text{ and } 4p} \quad \underbrace{=}_{\text{is}} \quad \underbrace{23}_{23}$
Simplify.	$p=23$
Check: $5p-4p=23$ $5(23)-4(23) \stackrel{?}{=} 23$ $115-92 \stackrel{?}{=} 23$ $23=23$ ✓	
The solution checks.	

TRY IT 8.19 Translate and solve: The difference of $4x$ and $3x$ is 14.

TRY IT 8.20 Translate and solve: The difference of $7a$ and $6a$ is -8.

Translate and Solve Applications

In most of the application problems we solved earlier, we were able to find the quantity we were looking for by simplifying an algebraic expression. Now we will be using equations to solve application problems. We'll start by

restating the problem in just one sentence, assign a variable, and then translate the sentence into an equation to solve. When assigning a variable, choose a letter that reminds you of what you are looking for.

EXAMPLE 8.11

The Robles family has two dogs, Buster and Chandler. Together, they weigh 71 pounds.

Chandler weighs 28 pounds. How much does Buster weigh?

Solution

Read the problem carefully.	
Identify what you are asked to find, and choose a variable to represent it.	How much does Buster weigh? Let b = Buster's weight
Write a sentence that gives the information to find it.	Buster's weight plus Chandler's weight equals 71 pounds.
We will restate the problem, and then include the given information.	Buster's weight plus 28 equals 71.
Translate the sentence into an equation, using the variable b.	$b + 28 = 71$
Solve the equation using good algebraic techniques.	$b + 28 - 28 = 71 - 28$ $b = 43$
Check the answer in the problem and make sure it makes sense.	
Is 43 pounds a reasonable weight for a dog? Yes. Does Buster's weight plus Chandler's weight equal 71 pounds?	
$43 + 28 \stackrel{?}{=} 71$	
$71 = 71$ ✓	
Write a complete sentence that answers the question, "How much does Buster weigh?"	Buster weighs 43 pounds

TRY IT 8.21 Translate into an algebraic equation and solve: The Pappas family has two cats, Zeus and Athena. Together, they weigh 13 pounds. Zeus weighs 6 pounds. How much does Athena weigh?

TRY IT 8.22 Translate into an algebraic equation and solve: Sam and Henry are roommates. Together, they have 68 books. Sam has 26 books. How many books does Henry have?

HOW TO

Devise a problem-solving strategy.

Step 1. Read the problem. Make sure you understand all the words and ideas.
Step 2. Identify what you are looking for.
Step 3. Name what you are looking for. Choose a variable to represent that quantity.

Step 4. Translate into an equation. It may be helpful to restate the problem in one sentence with all the important information. Then, translate the English sentence into an algebra equation.
Step 5. Solve the equation using good algebra techniques.
Step 6. Check the answer in the problem and make sure it makes sense.
Step 7. Answer the question with a complete sentence.

EXAMPLE 8.12

Shayla paid $\$24{,}575$ for her new car. This was $\$875$ less than the sticker price. What was the sticker price of the car?

Solution

What are you asked to find?	"What was the sticker price of the car?"
Assign a variable.	Let s = the sticker price of the car.
Write a sentence that gives the information to find it.	\$24,575 is \$875 less than the sticker price \$24,575 is \$875 less than s
Translate into an equation.	$24{,}575 = s - 875$
Solve.	$24{,}575 + 875 = s - 875 + 875$ $25{,}450 = s$
Check:	
Is \$875 less than \$25,450 equal to \$24,575?	
$25{,}450 - 875 \stackrel{?}{=} 24{,}575$	
$24{,}575 = 24{,}575$ ✓	
Write a sentence that answers the question.	The sticker price was \$25,450.

TRY IT 8.23 Translate into an algebraic equation and solve: Eddie paid $\$19{,}875$ for his new car. This was $\$1{,}025$ less than the sticker price. What was the sticker price of the car?

TRY IT 8.24 Translate into an algebraic equation and solve: The admission price for the movies during the day is $\$7.75$. This is $\$3.25$ less than the price at night. How much does the movie cost at night?

The *Links to Literacy* activity, "The 100-pound Problem", will provide you with another view of the topics covered in this section.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Solving One Step Equations By Addition and Subtraction (https://www.openstax.org/l/24Solveonestep)
Solve One Step Equations By Add and Subtract Whole Numbers (Variable on Left) (https://www.openstax.org/l/24SolveByAdd)
Solve One Step Equations By Add and Subtract Whole Numbers (Variable on Right) (https://www.openstax.org/l/24AddSubtrWhole)

SECTION 8.1 EXERCISES

Practice Makes Perfect

Solve Equations Using the Subtraction and Addition Properties of Equality

In the following exercises, determine whether the given value is a solution to the equation.

1. Is $y = \frac{1}{3}$ a solution of $4y + 2 = 10y$?

2. Is $x = \frac{3}{4}$ a solution of $5x + 3 = 9x$?

3. Is $u = -\frac{1}{2}$ a solution of $8u - 1 = 6u$?

4. Is $v = -\frac{1}{3}$ a solution of $9v - 2 = 3v$?

In the following exercises, solve each equation.

5. $x + 7 = 12$

6. $y + 5 = -6$

7. $b + \frac{1}{4} = \frac{3}{4}$

8. $a + \frac{2}{5} = \frac{4}{5}$

9. $p + 2.4 = -9.3$

10. $m + 7.9 = 11.6$

11. $a - 3 = 7$

12. $m - 8 = -20$

13. $x - \frac{1}{3} = 2$

14. $x - \frac{1}{5} = 4$

15. $y - 3.8 = 10$

16. $y - 7.2 = 5$

17. $x - 15 = -42$

18. $z + 5.2 = -8.5$

19. $q + \frac{3}{4} = \frac{1}{2}$

20. $p - \frac{2}{5} = \frac{2}{3}$

21. $y - \frac{3}{4} = \frac{3}{5}$

Solve Equations that Need to be Simplified

In the following exercises, solve each equation.

22. $c + 3 - 10 = 18$

23. $m + 6 - 8 = 15$

24. $9x + 5 - 8x + 14 = 20$

25. $6x + 8 - 5x + 16 = 32$

26. $-6x - 11 + 7x - 5 = -16$

27. $-8n - 17 + 9n - 4 = -41$

28. $3\,(y - 5) - 2y = -7$

29. $4\,(y - 2) - 3y = -6$

30. $8\,(u + 1.5) - 7u = 4.9$

31. $5\,(w + 2.2) - 4w = 9.3$

32. $-5\,(y - 2) + 6y = -7 + 4$

33. $-8\,(x - 1) + 9x = -3 + 9$

34. $3(5n - 1) - 14n + 9 = 1 - 2$

35. $2(8m + 3) - 15m - 4 = 3 - 5$

36. $-(j + 2) + 2j - 1 = 5$

37. $-(k + 7) + 2k + 8 = 7$

38. $6a - 5\,(a - 2) + 9 = -11$

39. $8c - 7\,(c - 3) + 4 = -16$

40. $8(4x + 5) - 5(6x) - x = 53$

41. $6(9y - 1) - 10(5y) - 3y = 22$

Translate to an Equation and Solve

In the following exercises, translate to an equation and then solve.

42. Five more than x is equal to 21.

43. The sum of x and -5 is 33.

44. Ten less than m is -14.

45. Three less than y is -19.

46. The sum of y and -3 is 40.

47. Eight more than p is equal to 52.

48. The difference of $9x$ and $8x$ is 17.

49. The difference of $5c$ and $4c$ is 60.

50. The difference of n and $\frac{1}{6}$ is $\frac{1}{2}$.

51. The difference of f and $\frac{1}{3}$ is $\frac{1}{12}$.

52. The sum of $-4n$ and $5n$ is -32.

53. The sum of $-9m$ and $10m$ is -25.

Translate and Solve Applications

In the following exercises, translate into an equation and solve.

54. Pilar drove from home to school and then to her aunt's house, a total of 18 miles. The distance from Pilar's house to school is 7 miles. What is the distance from school to her aunt's house?

55. Jeff read a total of 54 pages in his English and Psychology textbooks. He read 41 pages in his English textbook. How many pages did he read in his Psychology textbook?

56. Pablo's father is 3 years older than his mother. Pablo's mother is 42 years old. How old is his father?

57. Eva's daughter is 5 years younger than her son. Eva's son is 12 years old. How old is her daughter?

58. Allie weighs 8 pounds less than her twin sister Lorrie. Allie weighs 124 pounds. How much does Lorrie weigh?

59. For a family birthday dinner, Celeste bought a turkey that weighed 5 pounds less than the one she bought for Thanksgiving. The birthday dinner turkey weighed 16 pounds. How much did the Thanksgiving turkey weigh?

60. The nurse reported that Tricia's daughter had gained 4.2 pounds since her last checkup and now weighs 31.6 pounds. How much did Tricia's daughter weigh at her last checkup?

61. Connor's temperature was 0.7 degrees higher this morning than it had been last night. His temperature this morning was 101.2 degrees. What was his temperature last night?

62. Melissa's math book cost $\$22.85$ less than her art book cost. Her math book cost $\$93.75$. How much did her art book cost?

63. Ron's paycheck this week was $\$17.43$ less than his paycheck last week. His paycheck this week was $\$103.76$. How much was Ron's paycheck last week?

Everyday Math

64. **Baking** Kelsey needs $\frac{2}{3}$ cup of sugar for the cookie recipe she wants to make. She only has $\frac{1}{4}$ cup of sugar and will borrow the rest from her neighbor. Let s equal the amount of sugar she will borrow. Solve the equation $\frac{1}{4} + s = \frac{2}{3}$ to find the amount of sugar she should ask to borrow.

65. **Construction** Miguel wants to drill a hole for a $\frac{5}{8}$-inch screw. The screw should be $\frac{1}{12}$ inch larger than the hole. Let d equal the size of the hole he should drill. Solve the equation $d + \frac{1}{12} = \frac{5}{8}$ to see what size the hole should be.

Writing Exercises

66. Is -18 a solution to the equation $3x = 16 - 5x$? How do you know?

67. Write a word sentence that translates the equation $y - 18 = 41$ and then make up an application that uses this equation in its solution.